Birla Institute of Technology and Science, Pilani
Second Semester 2023–24
EEE F111: Electrical Sciences
Comprehensive Examination (Closed Book)

Maximum Marks: 135 **Time: 3 hrs** **Date: 15/05/2024**

Attempt all parts of a question in sequence. Full credit will only be given if the solution is neat and shows all the required steps.

1(A) Draw the I-V characteristics of a real and ideal PN junction silicon diodes. If a silicon diode is forward-biased at 20 mA and 0.8 V at 300 K, what will be the value of the saturation current of the diode at 300 K? **[10 M]**

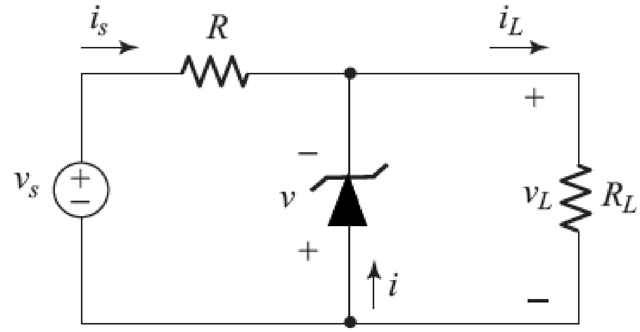

Figure 1(B)

1(B) A Zener diode is used in the circuit shown in the Figure 1(B). Identify an application of this circuit. Assume that the diode has a breakdown voltage of 7 V and a maximum and minimum current rating of 50 mA and 5 mA, respectively. If supply voltage is 28 V and R is 300 Ω, determine the range of R_L for which the diode will operate in the breakdown state. **[10 M]**

1(C) In the circuit shown in Figure 1(C), the BJTs Q_1 and Q_2 have dc betas h_{FE1} and h_{FE2}, respectively. Suppose that R_B = 68 kΩ, R_{C1} = 1 kΩ, R_{C2} = 500 Ω, R_E = 210 Ω, and V_{CC} = 5 V. Find the minimum values of h_{FE1} and h_{FE2} such that both Q_1 and Q_2 are in saturation. **[15 M]**

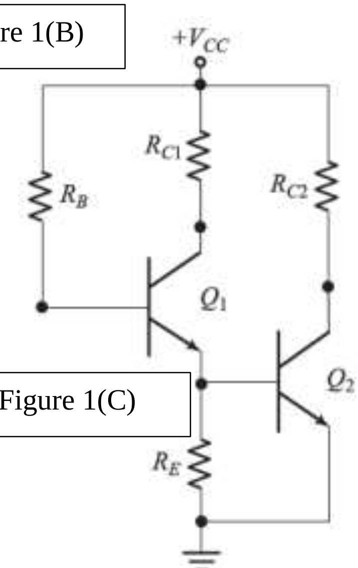

Figure 1(C)

2(A) In the ideal OPAMP circuit shown in Figure 2(A), the current flowing in 3R resistor is 'i'. Find the voltage gain (V_o/V_{in}). **[10 M]**

2(B) A three-phase circuit is shown in Figure 2(B). It has three voltage sources (rms), $\mathbf{V_1}$=110 $\angle 0^0$ V, $\mathbf{V_2}$= 110 $\angle 120^0$ V, $\mathbf{V_3}$=110 $\angle 240°$ V and three loads R=50 Ω, C= 1/60 F and L=15 H. The angular frequency is 3 rad/sec. Calculate (i) the current in the neutral wire (I_N) (ii) the real power consumed by circuit and (iii) reactive power consumed by circuit. **[20 M]**

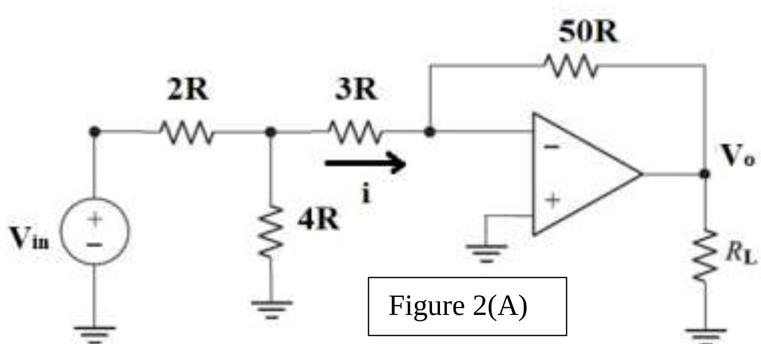

Figure 2(A)

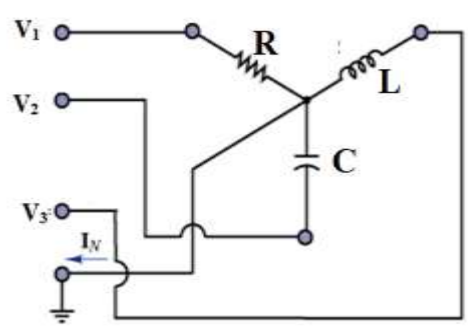

Figure 2(B)

PTO

3(A) For the filter circuit shown in Figure 3(A) find,

(i) The transfer function $\frac{\mathbf{V_o}(j\omega)}{\mathbf{V_i}(j\omega)}$
(ii) Sketch and properly label the magnitude response of the transfer function
(iii) The half-power frequency
(iv) Type of the filter **[15 M]**

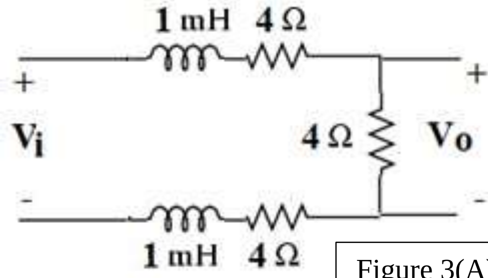

Figure 3(A)

3(B) As shown in Figure 3(B), a ring-shaped iron core (μ_r = 800, $\mu_0=4\pi\times10^{-7}$ H/m) has mean diameter of 50 cm and circular cross-section of radius 5 cm. As shown the ring includes an air-gap of 2 mm. A copper coil with 1000 turns is wound over the ring to produce a flux of 10 mWb in the ring. Neglect flux leakage and fringing effect in the core and the air gap. Calculate the following. **[20 M]**

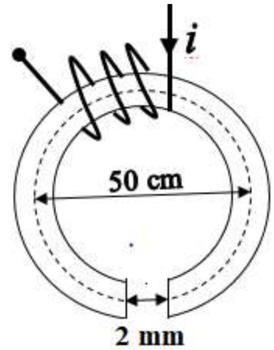

Figure 3(B)

(i) Flux density (B) in the ring
(ii) Current in the coil
(iii) Magnetic field intensity (H) in both the core and air-gap
(iv) Draw the analogous electrical circuit of the given magnetic circuit
(v) If the core is assumed to have infinite permeability, calculate the flux by using the result of (ii)

4(A) Figure 4(A) shows the circuit using depletion MOSFETs M_1 which is operated in the ohmic region and M_2 operated in the active region. Calculate the voltages and currents marked (v_{GS1}, v_{GS2}, v_{DS1}, v_{DS2}, i_{D1}, and i_{D2}) and confirm the regions of operations for M_1 and M_2. Given for M_1, I_{DSS}=4 mA, V_p = –4 V and for M_2, I_{DSS}=8 mA, V_p = –4 V **[15 M]**

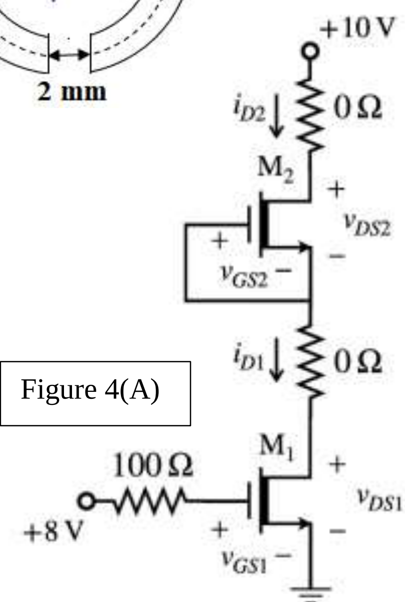

Figure 4(A)

Given $i_D = I_{DSS}\left(1-\frac{v_{GS}}{V_P}\right)^2 \; or \; I_{DSS}\left[2\left(1-\frac{v_{GS}}{V_P}\right)\frac{v_{DS}}{-V_P}-\left(\frac{v_{DS}}{V_P}\right)^2\right]$

4(B) In the transformer circuit shown in Figure 4(B), if the angular frequency of operation is 314 rad/sec, calculate the self-inductances of the primary and secondary windings, mutual inductance between them, and the coupling coefficient (k). State whether this transformer is ideal or not. Using frequency domain analysis, calculate the phasors I_1, I_2, impedance seen by V_s, complex power supplied by V_s, and the complex power received by the load Z_L. **[20 M]**

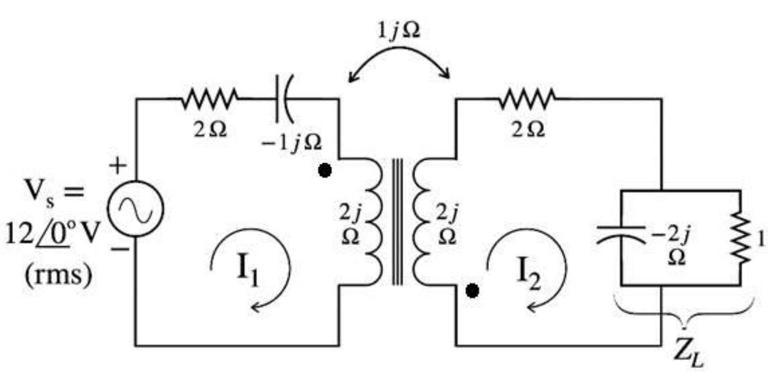

Figure 4(B)

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